16 (a) Use the iteration formula $x_{n+1} = \sqrt[3]{10 - 2x_n}$ to find the values of x_1, x_2 and x_3
Start with $x_0 = 2$

$x_1 = \text{cuberoot}(10 - 2x_0) = \text{cuberoot}(6) = 1.8171...$

$x_2 = \text{cuberoot}(10 - 2x_1) = 1.8533...$

$x_3 = \text{cuberoot}(10 - 2x_2) = 1.8462...$

1 mark
correct x1 substitution

1 mark
iterate x2 then x3

1 mark
all three values

$x_1 =$ 1.817

$x_2 =$ 1.853

$x_3 =$ 1.846

(3)

The values of x_1, x_2 and x_3 found in part (a) are estimates of the solution of an equation of the form $x^3 + ax + b = 0$ where a and b are integers.

(b) Find the value of a and the value of b.

At a fixed point, $x = \text{cuberoot}(10 - 2x)$.

Cube: $x^3 = 10 - 2x$, so $x^3 + 2x - 10 = 0$.

Therefore $a = 2$ and $b = -10$.

1 mark
a=2, b=-10

$a =$ 2

$b =$ -10

(1)

(Total for Question 16 is 4 marks)

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